AI-500: Foundations of Artificial Intelligence

Assignment 2
Data Collection, Curation, Cleaning, Governance, and Exploratory Data Analysis

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Assignment Overview

This assignment follows a single real-world case study, *MissedCare*, from raw records to a modelling-ready dataset.

In the theoretical section, you will examine how a healthcare dataset is constructed. You will diagnose data-quality problems and missing-data mechanisms, detect data leakage in a modelling pipeline, and carry out short quantitative analyses of outliers, preprocessing contamination, evaluation metrics, and fairness.

In the programming section, you will work with the real MissedCare dataset in Python. You will assess its structure and quality, explore distributions and relationships through visualization, compare appointment attendance across patient subgroups, engineer temporal features, and build a leakage-free preprocessing pipeline that prepares the data for machine learning.

Learning Objectives

Upon successful completion, you will be able to:

- Explain how a dataset is constructed through definitions, measurement choices, sampling decisions, and institutional practices, and judge whether it is fit for a given purpose.
- Identify data-quality problems and classify them by quality dimension (completeness, accuracy, consistency, timeliness, representativeness, and measurement error).
- Distinguish between MCAR, MAR, and MNAR missingness and choose a suitable handling strategy for each.

- Recognize target leakage, train–test contamination, temporal leakage, group leakage, and site leakage, and design an evaluation that avoids them.
- Perform an end-to-end exploratory data analysis, including subgroup and intersectional analysis, using appropriate visualizations and descriptive statistics.
- Engineer meaningful features and prepare a dataset for modelling using preprocessing that is fitted on the training data only.

How to Complete This Assignment

A quick word before you begin: this document may look long, but please do not be put off. Most of its length comes from the scenario, dataset description, and instructions included to help you; the actual amount of work is much smaller than the page count suggests.

This assignment comes as two files on the LMS:

1. **This document (PDF).** It contains the scenario, all questions for Section 1 and Section 2, the marks for each question, the grading rubric, and the instructions. It does not contain any space for answers.
2. **The answer notebook,** `AI500_Assignment2_Answer_Template.ipynb`. It provides a designated space for every answer: Markdown cells for written answers in Section 1 and Section 2, and code cells for the programming work in Section 2.

Work through the assignment in the following order:

1. **Read this document in full first.** Make sure you understand the scenario, the dataset, and what each question asks before you write any answers.
2. **Then open the answer notebook** and work through it from top to bottom, starting with Section 1 (Part A in the notebook) and then Section 2 (Part B in the notebook).
3. **For each question, read it here first, then in the notebook.** Before answering any question, read its full statement in this document, then read the same question in the notebook, along with its answer format and, for Section 1, its **Read** note, before you start writing.

All of your work must be submitted in the answer notebook. This includes

your written answers to Section 1, your calculations and working, your code, your plots, and your written observations for Section 2. Do not submit answers in any other file or format.

Follow the assignment closely. Answer every question in the cell provided for it, keep to the requested format (for example, tables, one-line answers, or maximum sentence counts), and do not delete, reorder, or reword any question, table, or heading in the notebook.

Reading guidance. In the answer notebook, every Section 1 question (Part A) is accompanied by a **Read** note that lists the sections and page numbers of the Module 2 notes that you should read to answer that question. Use these notes before attempting each question. For Section 2 (Part B), the relevant material is in the Module 3 notes.

Section 1: Theory (Data Collection, Curation, Cleaning, and Governance) — 50 Points

Section 1 Deliverables

Write your answers to the six problems below in **Part A** of the answer notebook (`AI500_Assignment2_Answer_Template.ipynb`). The notebook provides a designated cell for each answer and states which sections of the notes to read for each question. All six problems are based on the MissedCare scenario described in the box below; read it carefully before you begin. The relevant theory is covered in the Module 2 notes (*Data Collection, Curation, Cleaning and Governance*) provided on LMS.

The MissedCare Scenario

The public health clinics of Vitória, a city in the state of Espírito Santo, Brazil, lose roughly one appointment in every five to patients who book an appointment but do not turn up. Every empty slot is clinical time that could have been given to another patient who needed care.

The municipal health department proposes the following project:

> **"Build an AI model that runs every evening and produces a list of the patients who are most likely to miss their appointment the next day, so that clinic staff can phone**

them first."

The department has access to the clinics' appointment records for April to June 2016: around 110,000 appointments made by about 62,000 patients. Each record describes one appointment. It includes the patient's identifier, gender, and age; the neighbourhood of the clinic; whether the patient is enrolled in *Bolsa Família* (a government welfare programme for low-income families); whether the patient has hypertension, diabetes, alcoholism, or a disability; whether an SMS reminder was sent; the date and time the appointment was booked; the date of the appointment itself; and whether the patient attended.

In the dataset, the attendance column is named `No-show`. A value of `"Yes"` means the patient **missed** the appointment, and a value of `"No"` means the patient attended.

You are the data scientist on this project. Before anyone trains a model, your job is to establish whether the data can be trusted and what it can and cannot be used for.

Problem 1: Constructing the Dataset (8 Points)

A dataset is not simply found; it is produced through a series of decisions. Using the MissedCare scenario, work through the following steps.

1. **Defining the target (2 Points).** Write three different operational definitions of a "no-show". For a patient who arrives 40 minutes late for the appointment, state the label (no-show or attended) that the patient would receive under each of your three definitions.

2. **Choosing variables (2 Points).** Propose four variables you would include in the dataset and two variables you would deliberately exclude. For each variable, give a one-line reason based on relevance, privacy, or reliability.

3. **Sampling (1 Point).** The records come only from public clinics in a single city. Identify who is missing from this dataset and describe one deployment setting in which this gap would matter.

4. **Interpreting "NA" (1 Point).** Suppose the column `SMS_received` is blank for some appointments. Describe three different real-world situations that the blank could represent.

5. **Same name, different meaning (1 Point).** Clinic A records a no-show

when a patient is more than 15 minutes late. Clinic B records a no-show only when the patient never arrives that day. Explain what goes wrong if the two clinics' records are combined into a single training dataset.

6. **Fitness for purpose (1 Point).** For each of the following uses, state whether the dataset is fit for purpose and give a one-line reason:

   (i) a monthly attendance report;

   (ii) tomorrow's call list;

   (iii) a single national model for all of Brazil;

   (iv) a study claiming that SMS reminders *cause* patients to attend.

Problem 2: Synthetic Data (3 Points)

An external software vendor asks the health department for a *synthetic* copy of the MissedCare records instead of the real ones, to develop a scheduling product.

1. Identify two risks of sharing the synthetic copy. For each risk, name the validation dimension (fidelity, utility, privacy, fairness, coverage, or realism) that would detect it. (2 Points)

2. State whether the following claim is true or false, and justify your answer in one sentence: "Because the data are synthetic, they are anonymous." (1 Point)

Problem 3: Spotting Data-Quality Flaws (10 Points)

The raw Kaggle version of the MissedCare dataset reports **zero missing values** across more than 110,000 rows, so at first glance it appears clean. Table 1 shows twelve rows taken from a triage-desk export of the same records, to which a `Weight_kg` column has been added.

Table 1: Twelve rows from the MissedCare triage-desk export.

#	AppointmentID	PatientId	Gender	Age	Weight_kg	Neighbourhood	ScheduledDay	AppointmentDay	SMS	No-show
1	5642903	2.98725e+13	F	62	71	JARDIM DA PENHA	2016-04-29	2016-04-29	0	No
2	5642503	558997776694438	M	56	80	JARDIM DA PENHA	2016-04-29	2016-04-29	0	No
3	5642549	4262962299951	Female	8	27	MATA DA PRAIA	2016-04-29	2016-04-29	0	No
4	5642828	867951213174	f	N/A	64	PONTAL DE CAMBURI	2016-04-27	2016-04-29	0	No
5	5642494	8841186448183	F	34	154	JARDIM DA PENHA	2016-04-29	2016-04-29	0	No
6	5626772	95985133231274	F	76		REPÚBLICA	2016-04-27	2016-04-29	0	Yes
7	5630279	733688164476661	M	23	78	GOIABEIRAS	2016-04-27	2016-04-29	0	Yes
8	5630279	733688164476661	M	23	78	GOIABERAS	2016-04-27	2016-04-29	0	Yes
9	5633460	12154843752835	M	-1	3.2	CONQUISTA	2016-04-28	2016-04-29	0	No
10	5650093	78124564369297	F	115	58	ANDORINHAS	2016-05-03	2016-04-29	1	No
11	5634718	3449833394123	F	38	65	NOVA PALESTINA	2016-04-26	2016-04-29	1	1
12	5639264	2263865652	male	19	70	SANTO ANDRÉ	2016-04-26	2016-04-29	1	No

1. Identify every data-quality issue you can find in Table 1; there are at least twelve. For each issue, state the row(s) affected, describe the problem, name the data-quality dimension it violates (completeness, accuracy, consistency, timeliness, representativeness, or measurement error), and propose a fix. Present your answer as a table. (7 Points)

2. Row 9 records an age of -1 and a weight of 3.2 kg. Decide whether this row should be deleted as an error or decoded as a meaningful code, and justify your decision in one sentence. (1 Point)

3. Two receptionists independently coded the *reason* for 50 missed appointments as either "patient-side" or "clinic-side". They agreed on 40 of the 50 appointments. Receptionist 1 chose "patient-side" 30 times, and Receptionist 2 chose "patient-side" 28 times. Compute Cohen's κ, showing your working, and interpret the value. Then explain in one sentence why a majority vote across several receptionists would still not guarantee the true reason. (2 Points)

Problem 4: Diagnosing Missingness (9 Points)

For each of the three situations below, (a) name the missingness mechanism (MCAR, MAR, or MNAR), (b) propose a handling strategy, and (c) justify your choice in one sentence. Each situation carries 3 Points.

1. A server outage erased the SMS logs for three randomly timed afternoons.

2. The blood test `HbA1c` is recorded only when a doctor suspects that a patient's diabetes is poorly controlled. A teammate proposes filling the missing values with a k-nearest-neighbours (KNN) imputer. As part of your justification, explain why this proposal is a mistake.

3. `Neighbourhood` is blank for patients who have no fixed address.

Problem 5: Detecting Data Leakage (8 Points)

A teammate reports that the no-show model achieves an AUC of **0.97** and describes the pipeline as follows:

1. For each patient, a feature `past_noshow_rate` is computed as the proportion of **all** of that patient's appointments in the file that were missed.

2. A feature `reminder_call_outcome` is added. Clinic staff record this value on the day *after* the appointment.
3. `StandardScaler` and SMOTE are applied to the full dataset, and the model is then evaluated with 5-fold `KFold` cross-validation.
4. Many patients have ten or more appointments, and their records are spread across both the training and validation folds.
5. The model will next be deployed in a different city, but it has been tested only on appointments from Vitória's neighbourhoods.

1. For each of the five steps, identify the type of leakage involved (target leakage, train–test contamination, temporal leakage, group leakage, or site leakage) and propose a fix. A step may involve more than one type. (5 Points)
2. For each of the following evaluation setups, state which step(s) of the pipeline it addresses and why: `StratifiedGroupKFold`, `TimeSeriesSplit`, and a neighbourhood hold-out. (3 Points)

Problem 6: Quantitative Analysis (12 Points)

Show all of your working and round final answers to two decimal places.

Part A: Outliers in Waiting Time (3 Points)
The waiting times, in days, for eleven appointments are:

$$0,\ 0,\ 1,\ 2,\ 3,\ 4,\ 6,\ 7,\ 10,\ 14,\ 60$$

1. Compute Q_1, Q_3, the IQR, and the fences $Q_1 - 1.5\,\mathrm{IQR}$ and $Q_3 + 1.5\,\mathrm{IQR}$, using linear interpolation (as `numpy.percentile` does). Which values are flagged as outliers? (1.5 Points)
2. Using the sample standard deviation (divisor $n-1$), compute the z-scores of 14 and 60. Which values does the rule $|z| > 3$ flag? (1.5 Points)

Part B: Preprocessing Contamination (3 Points)
The ages of the patients in a training set are 20, 30, 40, and 50, and the ages in the test set are 80 and 90. Use the population standard deviation (divisor n), as `StandardScaler` does.

1. Standardize the test patient aged 90 using the mean and standard deviation of the training set only. (1.5 Points)

2. Standardize the same patient using the mean and standard deviation of all six patients. Explain in one sentence why this leaky value looks "tamer". (1.5 Points)

Part C: Why Accuracy Misleads (3 Points)

A test set contains 1,000 appointments, of which 200 were missed. The model flags 210 appointments as likely no-shows, and 120 of these are actually missed.

1. Compute the accuracy and recall of a baseline that always predicts "attended". Then compute the model's accuracy, precision, and recall. (2 Points)

2. Platt scaling maps a model score s to a probability using $P(\text{miss}) = \frac{1}{1+\exp(As+B)}$. With $A = -2$ and $B = 1.5$, compute $P(\text{miss})$ for a score of $s = 1$. (1 Point)

Part D: Fairness from Counts (3 Points)

The same test set, broken down by gender, is shown in Table 2.

Table 2: Model results on the test set by gender.

Group	Appointments	Missed	Flagged	Flagged and missed
Women	600	150	150	90
Men	400	50	60	30

1. For each group, compute the positive prediction rate, the true positive rate (TPR), the false positive rate (FPR), and the positive predictive value (PPV). (1.5 Points)

2. State which of the following fairness criteria hold: demographic parity, equal opportunity, equalized odds, and predictive parity. Give a one-line justification for each. (1.5 Points)

Section 2: Programming (End-to-End EDA and Data Preparation) — 50 Points

Section 2 Deliverables

Complete this section in **Part B** of the same answer notebook. Part B contains designated code cells for your code and outputs, and Markdown cells for your written answers, for each of the five problems below. The relevant material is cov-

ered in the Module 3 notes (*Exploratory Data Analysis, Visualization, Descriptive Statistics, and Feature Engineering*) and, for data leakage, in the Module 2 notes.

Scenario

You are continuing as the data scientist on the MissedCare project introduced in Section 1. The municipal health department of Vitória wants to understand why so many patients miss their appointments, and eventually wants a model that identifies, each evening, the patients most likely to miss the next day's appointment.

Before any model is built, you have been asked to deliver a thorough exploratory analysis and a modelling-ready dataset. Specifically, the department wants to know:

- what the data contain and how reliable they are;
- how patient characteristics and appointment timing relate to attendance;
- whether attendance differs across groups of patients;
- which features are likely to be useful for prediction; and
- how the data should be prepared so that a future model can be trained and evaluated fairly, without leakage.

The Dataset

You will use the **Medical Appointment No Shows** dataset, publicly available on Kaggle:

`https://www.kaggle.com/datasets/joniarroba/noshowappointments`

You need a free Kaggle account to download it. The download contains a single CSV file, `KaggleV2-May-2016.csv`.

Context. The data were collected from public health clinics in Vitória, Espírito Santo, Brazil. They cover medical appointments scheduled to take place between late April and early June 2016.

Size and structure. The file contains approximately 110,500 rows and 14 columns. **Each row represents one medical appointment**, not one patient. The same patient may appear in several rows if they booked more than one ap-

pointment during the period.

Target variable. The column `No-show` records whether the patient missed the appointment. Pay careful attention to its coding: `"Yes"` means the patient **did not attend**, and `"No"` means the patient **attended**.

Column names. Some column names contain spelling errors from the original source (for example, `Hipertension` and `Handcap`). They are listed below exactly as they appear in the file.

Table 3: Description of the variables in the MissedCare dataset.

Column	**Recorded as**	**Description**
`PatientId`	Number	Identification number of the patient. A patient can appear in several rows.
`AppointmentID`	Integer	Identification number of the appointment.
`Gender`	Text (F/M)	Gender of the patient.
`ScheduledDay`	Date and time	The date and time at which the appointment was booked (registered in the system).
`AppointmentDay`	Date	The date on which the appointment was due to take place. The time part of this field is always recorded as 00:00:00.
`Age`	Integer	Age of the patient, in years.
`Neighbourhood`	Text	The neighbourhood in which the appointment takes place.
`Scholarship`	0/1	1 if the patient is enrolled in *Bolsa Família*, a Brazilian government welfare programme for low-income families; 0 otherwise.
`Hipertension`	0/1	1 if the patient has hypertension; 0 otherwise.
`Diabetes`	0/1	1 if the patient has diabetes; 0 otherwise.
`Alcoholism`	0/1	1 if the patient has alcoholism; 0 otherwise.
`Handcap`	Integer	Disability indicator for the patient, recorded as an integer.
`SMS_received`	0/1	1 if one or more SMS reminders were sent to the patient for this appointment; 0 otherwise.
`No-show`	Text (Yes/No)	**Target.** `"Yes"` if the patient **missed** the appointment; `"No"` if the patient attended.

Problem 1: Understanding and Assessing the Dataset (10 Points)

Load the MissedCare dataset and perform an initial assessment of its structure and quality.

(a) **Dataset structure and quality (4 Points).** Report the number of observations and variables. Identify the numerical, categorical, binary, and temporal variables, and investigate missing values and duplicate records.

(b) **Descriptive statistics (4 Points).** Generate appropriate descriptive statistics for the numerical variables. For `Age`, additionally report the minimum, Q_1, median, Q_3, maximum, and IQR.

(c) **Initial observations (2 Points).** Based on your analysis, identify at least two observations about the structure or quality of the dataset that should be considered before proceeding with further analysis.

Problem 2: Exploring Distributions and Relationships (10 Points)

Use appropriate visualizations to investigate the distributions of important variables and their relationships with the target variable, `No-show`. Your analysis must include:

- the distribution of `Age`;
- the distribution of at least two relevant categorical or binary variables;
- the relationship between `Age` and `No-show`; and
- a correlation heatmap for appropriate numerical and binary variables.

For each visualization, briefly explain why the visualization is appropriate and report at least one meaningful observation from the resulting plot.

Problem 3: Investigating Appointment No-Shows Across Subgroups (10 Points)

The MissedCare team wants to understand whether appointment attendance differs across different groups of patients.

(a) **Patient characteristics (3 Points).** Calculate the overall no-show rate and investigate how it varies across at least four relevant patient characteris-

tics. Your analysis must include `Gender` and `SMS_received`. Use appropriate visualizations to compare the groups.

(b) **Age groups (2 Points).** Divide patients into meaningful age groups and compare their no-show rates.

(c) **Intersectional analysis (3 Points).** Perform an intersectional analysis of `Gender` $\times$ Age Group and visualize the resulting no-show rates.

(d) **Interpretation (2 Points).** Based on your analysis, describe the most notable differences you observe between groups. Explain whether examining the intersection of two characteristics reveals anything that may not be apparent when looking at each characteristic individually.

Problem 4: Temporal Analysis and Feature Engineering (10 Points)

The dataset contains information about when an appointment was scheduled and when it was supposed to take place.

(a) **Temporal features (3 Points).** Convert `ScheduledDay` and `AppointmentDay` into appropriate datetime formats and create meaningful temporal features, such as the day of the week, the month, and the scheduled hour.

(b) **Waiting time (3 Points).** Create a new feature, `WaitingDays`, representing the number of days between the scheduled date and the appointment date. Investigate the distribution of `WaitingDays` and examine how the no-show rate changes across different waiting-time groups.

(c) **Temporal patterns (2 Points).** Investigate whether the no-show rate varies across days of the week or months.

(d) **Feature usefulness (2 Points).** Based on your analysis, identify the temporal or engineered feature(s) that appear potentially useful for predicting appointment no-shows, and briefly explain why.

Problem 5: Preparing the Dataset for Modelling (10 Points)

Prepare the dataset for use in a machine-learning model that predicts whether a patient will miss an appointment. Your preprocessing should:

(a) encode the target variable `No-show` into an appropriate numerical representation (1 Point);

(b) select the features you would use for modelling and briefly justify any important exclusions (2 Points);

(c) split the data into training and test sets (1 Point);

(d) apply one-hot encoding to appropriate categorical variables (1 Point);

(e) apply Min–Max normalization to appropriate numerical features (1 Point);

(f) organize the preprocessing so that all transformations are fitted only on the training data and then applied to the test data (1 Point); and

(g) show the shape and a sample of the resulting modelling-ready training and test datasets (1 Point).

Finally, explain why fitting preprocessing transformations on the complete dataset before splitting it into training and test sets can lead to data leakage. (2 Points)

Execution & Formatting Instructions:

- Write your code in the designated code cells, and your explanations and observations in the designated Markdown cells. You may add extra code cells beneath a problem if needed. For each question, provide the required code, its output or visualization, and a brief interpretation.
- Ensure your full name and roll number are entered in the designated cell at the top of the notebook.
- Base your interpretations on the results you obtain, not on generic statements.
- Every plot must have a title, labelled axes, and a legend where needed. Where a comparison depends on group size, report the number of observations.
- Load the raw data once and work on a copy; do not overwrite the original file.
- Use only `pandas`, `numpy`, `matplotlib`, `seaborn`, and `scikit-learn`. Where randomness is involved, set `random_state = 42`.
- Before submission, you **must** execute your notebook from scratch to ensure there are no hidden state errors. Restart your environment and run all cells sequentially.

- Verify that all required outputs, tables, and plots are clearly visible in your final output before saving.

Grading Rubric

Criterion	Weight	Max Points
Section 1: Theory	**50%**	**50**
P1: Constructing the Dataset	**8%**	**8**
Target definitions & relabelling	-	2
Variable inclusion & exclusion	-	2
Sampling, "NA", & same-name issues	-	3
Fitness for purpose	-	1
P2: Synthetic Data	**3%**	**3**
Risks & validation dimensions	-	2
Anonymity claim	-	1
P3: Spotting Data-Quality Flaws	**10%**	**10**
Issues, dimensions & fixes	-	7
Row 9 decision	-	1
Cohen's κ & interpretation	-	2
P4: Diagnosing Missingness	**9%**	**9**
Mechanism, strategy & justification (3 cases)	-	9
P5: Detecting Data Leakage	**8%**	**8**
Leakage types & fixes	-	5
Evaluation setups	-	3
P6: Quantitative Analysis	**12%**	**12**
Outliers in waiting time	-	3
Preprocessing contamination	-	3
Why accuracy misleads	-	3
Fairness from counts	-	3
Section 2: Programming	**50%**	**50**
P1: Understanding & Assessing the Dataset	**10%**	**10**
Dataset structure & quality	-	4
Descriptive statistics	-	4
Initial observations	-	2
P2: Distributions & Relationships	**10%**	**10**
Distribution of Age	-	2.5
Categorical / binary distributions	-	2.5
Age vs. No-show	-	2.5
Correlation heatmap	-	2.5

Continued on the next page

Criterion	Weight	Max Points
P3: Subgroup Analysis	**10%**	**10**
Patient characteristics	-	3
Age groups	-	2
Intersectional analysis	-	3
Interpretation	-	2
P4: Temporal Analysis & Feature Engineering	**10%**	**10**
Temporal features	-	3
Waiting time	-	3
Temporal patterns	-	2
Feature usefulness	-	2
P5: Preparing for Modelling	**10%**	**10**
Target encoding & feature selection	-	3
Train–test split, one-hot encoding & Min–Max normalization	-	3
Training-only fit & modelling-ready datasets	-	2
Data leakage explanation	-	2
Total	**100%**	**100**

Submission Requirements

Submit **one file only** to the LMS: your completed answer notebook, containing all of your work for both Section 1 (Part A) and Section 2 (Part B). Name this file: `AI500_A2_<RollNumber>.ipynb`

Submission Notes:

Due: **October 13, 2026, 11:59 PM PKT**
Total Points: **100** (Section 1: 50 Points and Section 2: 50 Points)

- Do not submit a separate PDF report or any other file. Answers written outside the notebook will not be graded.
- Enter your full name and roll number in the designated cell at the top of the notebook.
- Before submitting, restart the kernel and run all cells from top to bottom (Kernel → Restart & Run All), so that every output, table, and plot is visible in the submitted notebook.
- Do not compress the notebook into a ZIP file.

Academic Integrity

- All code and written responses must be your own work.
- Collaboration on conceptual understanding is encouraged, but implementation must be individual.
- The use of generative AI tools is governed by the course AI policy stated in the course outline.

Best of luck, and remember that any help you ask for will be given, so reach out whenever you are stuck, and may your data be clean and your plots tell the truth!

— End of Assignment —